

Phantom (m:phant) Spec Showcase

Expression: $a + [\text{phantom}(\text{xyz})] + b$

Case 1: No m:phantPr (bare m:phant)

$$a + xyz + b$$

Case 2: Empty m:phantPr

$$a + xyz + b$$

Case 3:

1. ~~Definite integral (baseline — msubsup):~~

$$\int_{\theta}^1 f(x) dx$$

2. ~~Word's indefinite integral (uses subHide/supHide m:val=1):~~

$$\int f(x) dx$$

3. ~~Summation default limLoc — Finding A trigger:~~

$$\sum_{i=1}^n i$$

4. ~~Summation explicit limLoc=subSup (control, different vars):~~

$$\sum_{j=\theta}^m j$$

5. ~~Product default limLoc — Finding A:~~

$$\prod_{k=1}^N k$$

6. ~~Union default limLoc — Finding A (also supHide from Word):~~

$$\bigcup_i A_i$$

2b. ~~Crafted indefinite integral — NO m:sub/m:sup elements (Finding C real trigger):~~

$$\int_{-}^{+} f(x) dx$$

7. ~~subHide m:val="true" (Finding B==anchors-commit 2bd58d3):~~

$$\int_{-}^{+01} f$$

8. Bare ~~subHide~~ (Finding B==spec bare-flag=ON):

$$\int_{-}^{+01} f$$

9. ~~limLoc-present-no-val (Finding A==spec default=undOvr):~~

$$\sum_{i=1}^n i$$

10. ~~chr-present-no-val (Finding D==empty character):~~

$$\frac{b}{a}x$$

12. ~~subHide m:showval="0" (ST_OnOff OFF value==§22.9.2.7):sub
should stay visible→msubsup~~

$$a + xyz + b$$

Case 4: m:show val=1 (visible)

$$a + xyz + b$$

Case 5: m:show val=0 (hidden)

$$a + \quad + b$$

Case 6: m:show val=true

$$a + xyz + b$$

Case 7: m:show val=false (hidden)

$$a + \quad + b$$

Case 8: zeroWid alone

Case 9: zeroAsc alone

$$a + xyz + b$$

Case 10: zeroDesc alone (classic smash)

$$a + xyz + b$$

Case 11: show=0 + zeroDesc=1 (hidden, smashed)

$$a + \quad + b$$

Case 12: all three zeroed, show absent

Case 13: all three zeroed + show=0

$$a + \quad + b$$

Case 14: m:transp=1 (unsupported)

$$a + xyz + b \int_{\vartheta}^{\pm} f$$

~~13. m:grow m:val="0" (Finding E — §22.1.2.72): operator should NOT grow (expect largeop/stretchy=false on mo)~~

$$\sum_{i=1}^{\#} i$$